MSc THESIS · ARTEFACT PRESENTATION AND DEFENCE

Measuring Technical Debt in Mission-Critical Trading Systems

A Service-Level Quantitative Framework

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THE PROBLEM

Debt that source-code tools cannot see

In mission-critical trading IT, technical debt accumulates **outside source code:** in infrastructure-lifecycle exposure, patch delay and operational instability.

- Dominant tools (SonarQube, SQALE, CAST) operationalise debt from source-code rule violations. They are inapplicable where code is vendor-hosted or outsourced.
- Even with code access, three tools on the same codebase can return substantially divergent estimates (Amanatidis *et al.*, 2020).
- Infrastructure, platform and vulnerability debt stay largely invisible to static analysis.

4.9× higher odds of compromise at just one month of patch delay (Tizio, Armellini and Massacci, 2023)

84% fewer incidents after targeted architectural-debt repayment (de Toledo *et al.*, 2021)

16% drop in return on assets per 10% rise in debt; more than doubling within six years (Banker, Liang and Ramasubbu, 2021)

The gap: *no quantitative, service-level way to prioritise debt where source code cannot be reached.*

Four fields meet — none joins them up

Across 44 prioritisation studies, Lenarduzzi *et al*. (2021) find quantitative multi-criteria approaches uncommon and industrial validation rare.

	Service-level operational indicators	Multi-criteria prioritisation	Live operational data (ITSM/CMDB)	Regulated trading-IT context
Static-analysis tools (Biazotto *et al.*, 2024; Amanatidis *et al.*, 2020)	–	–	–	–
TD prioritisation (Lenarduzzi *et al.*, 2021; Alfayez *et al.*, 2023)	–	✓	–	–
DORA metrics as proxies (Forsgren, Humble, and Kim, 2018; Nord, Ozkaya and Woody, 2021)	~	–	~	–
TD in financial services (Haki *et al.*, 2023; Rolland and Lyytinen, 2021)	~	–	–	✓
TOPSIS / MCDM in IT (Albarak *et al.*, 2022; Khan and Purohit, 2022)	–	✓	–	–
This thesis: TOPSIS on live ITSM / CMDB data	✓	✓	✓	✓

✓ addressed ~ partial – not addressed

What the study set out to answer

Aim: Develop a service-level quantitative framework that prioritises technical debt in mission-critical trading systems from operational service-management data, not source-code analysis.

RQ1 PRIMARY

How can technical debt in mission-critical trading systems be prioritised quantitatively, at the service level, using operational data?

RQ2

To what extent do operational service-management metrics serve as valid proxies for technical-debt severity in vendor-dominated enterprise IT?

RQ3

How stable are the resulting service rankings when the criterion weights are perturbed?

A Design-Science artefact, evaluated on real data

Paradigm — Pragmatism

A knowledge claim is judged by whether the artefact works in the context it was built for. Debt is treated as a construct with no single true measurement.

Strategy — Design Science Research

Hevner *et al.* (2004) seven guidelines; Peffers *et al.* (2007) six-step process. The contribution is a functional artefact, evaluated analytically.

Case — Embedded single-organization

Trading-IT division of a systemically important Swiss financial institution. Unit of analysis: the IT service (a CMDB configuration item). Cross-sectional, Jan 2024 – Dec 2025.

Ethics & governance

Anonymised at source; no personal data; Head-of-Trading-IT authorisation. nFADP; BCS Code of Conduct. Scores describe services, not teams.

THE INSTRUMENT

Five operational indicators, mapped to debt dimensions

#	Indicator	Debt dimension	Source	Direction
1	**Incident frequency**	*Architectural debt*	ITSM	cost
2	**Mean time to recover (MTTR)**	*Infrastructure / platform debt*	ITSM	cost
3	**Change failure rate (CFR)**	*Process debt*	ITSM	cost
4	**Patch recency**	*Vulnerability debt*	CMDB	benefit
5	**Unsupported component-months / component**	*Infrastructure-lifecycle debt*	CMDB	cost

3
ITSM

2
CMDB

Candidate proxies, not established substitutes

Excluded on data availability: documentation completeness, test-coverage ratio and architectural coupling. All need source repositories or CI/CD artefacts unavailable for vendor-managed components.

TOPSIS — one procedure from data to ranked priority

Operational data	Five indicators	TOPSIS core	Ranked priority
ITSM incidents & changes · CMDB components & patch dates	Normalise per-component · label cost / benefit	Vector-normalise → weight → ideal best / worst → distances	Closeness $C_i \in [0,1]$ → ordered remediation list

Absolute-mode variant fixed bounds → rank-reversal-proof *(García-Cascales and Lamata, 2012; Yang, 2020)*

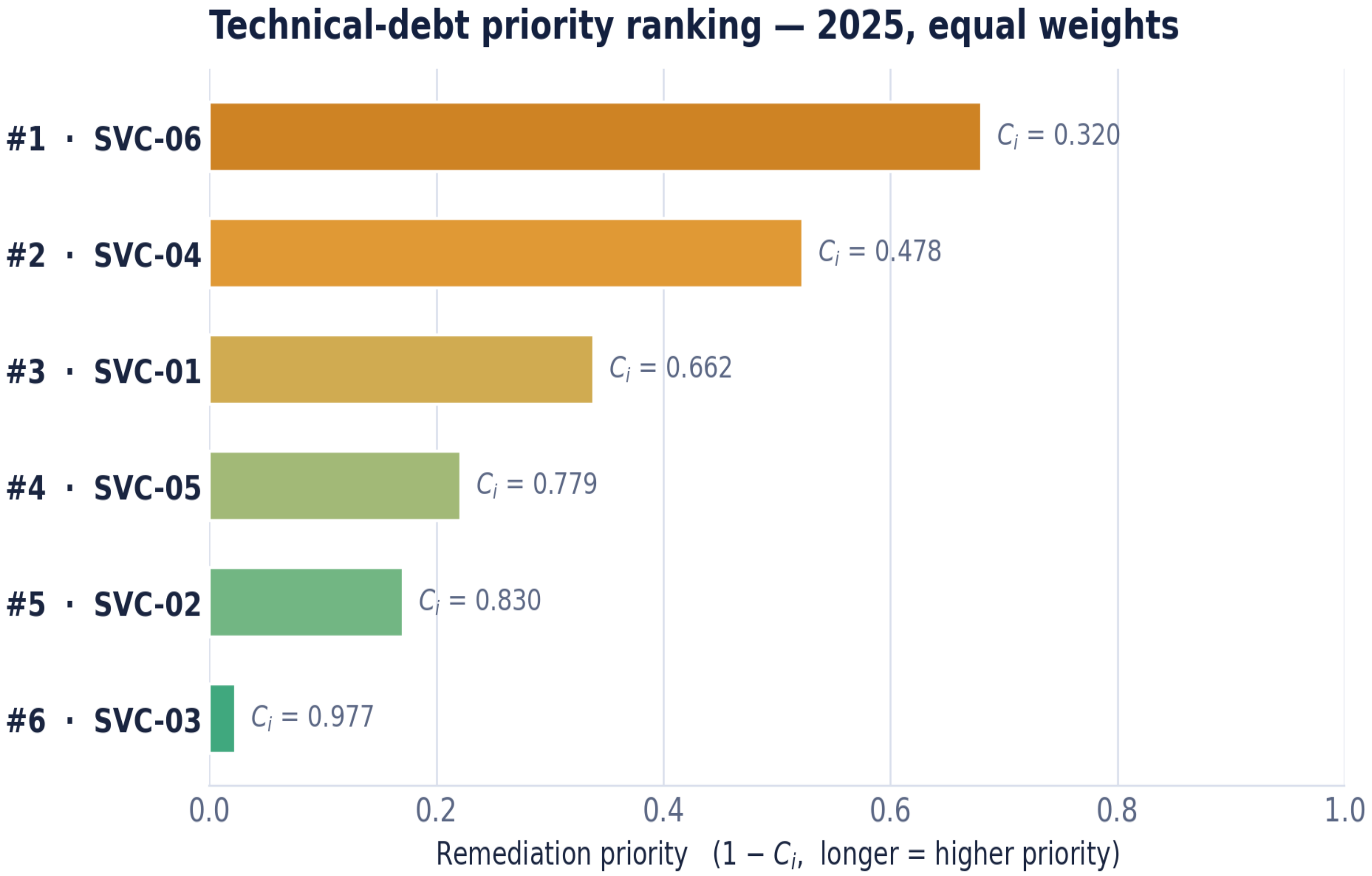
Why TOPSIS

- Integrates normalisation, weighting and ranking in one procedure (Hwang and Yoon, 1981).
- Handles mixed cost & benefit criteria directly, well suited to a ranked service portfolio.
- Robust to rank reversal among methods that exhibit it (Zanakis *et al.*, 1998).

Two transparent weightings

Equal (baseline): $w_j = 0.20$ each, the standard default without empirically justified weights.
Differential: MTTR & patch recency = 0.30; others = $0.1\overline{3}$. A practitioner judgement, stated as an explicit assumption and stress-tested in Chapter 5.

One consistent ranking — and it is not the obvious one



What the ranking says

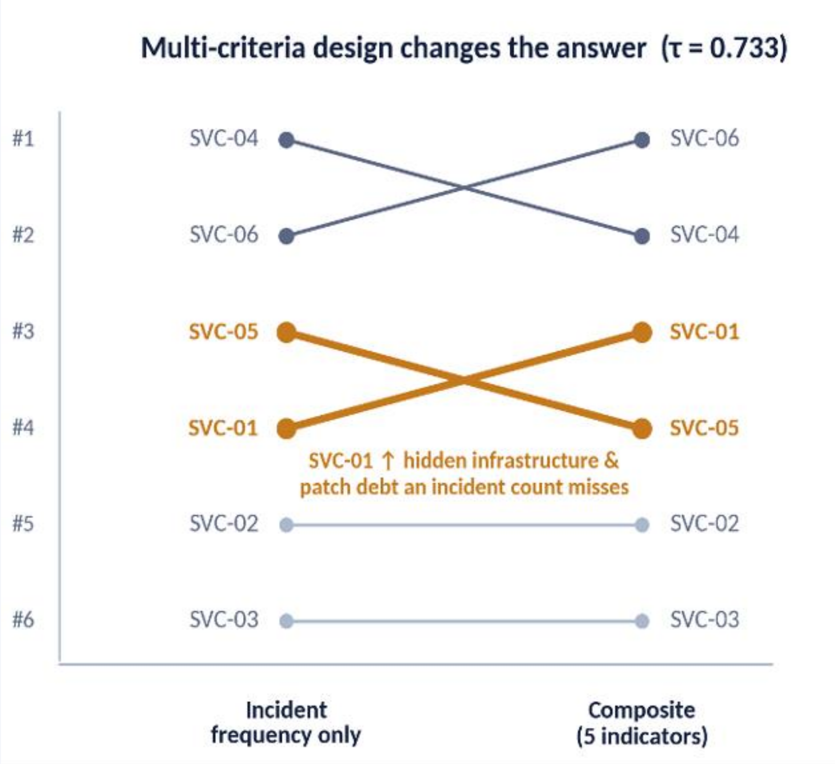
SVC-06 is top priority under both weightings and both years, worst or near-worst on four of five indicators.

SVC-03 is the invariant lowest, the well-maintained reference service.

Size-normalisation matters. On a raw cumulative measure, SVC-01's large estate would top the list. Per-component intensity correctly subordinates it to compact SVC-06, about half out of support, and for far longer.

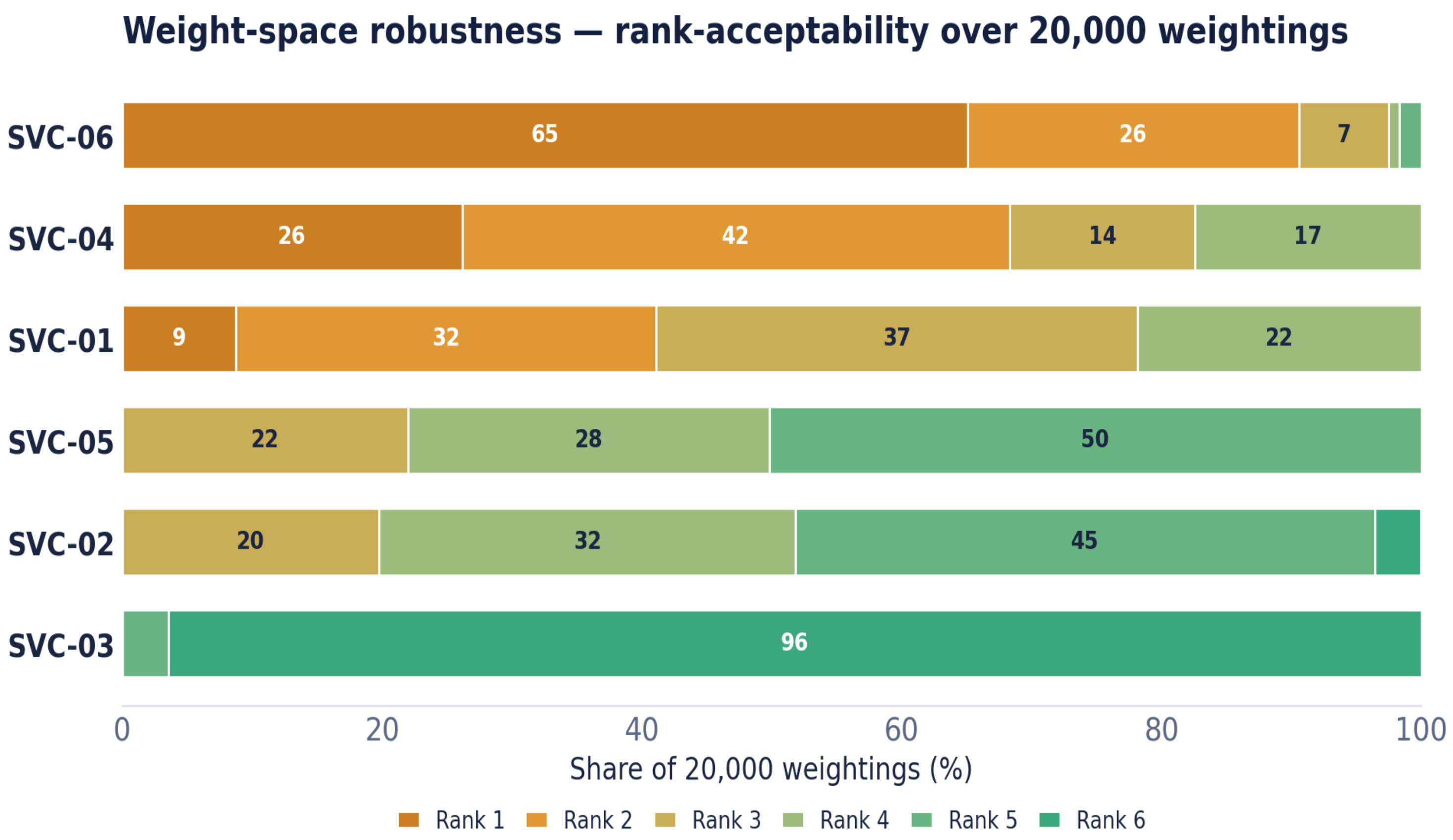
Before the order is trusted, it is tested on five fronts

1 Stability	Order survives weight change (τ = 0.867, 2025) and holds across years; the bottom rank never moves.
2 Convergent validity	SAW and VIKOR reproduce the order, absolute-mode agrees perfectly (τ = 1.000, 2025 equal).
3 Discriminant value	Diverges from an incident-only ranking (τ = 0.733) → it catches debt a single signal misses.
4 Structural rigour	Absolute-mode order holds under every one of six leave-one-out removals, in both years.
5 Weight-space	20,000 Monte-Carlo weightings, the acid test for RQ3 (next slide).



Front 3: two pairs swap, but SVC-01's rise is the telling one.

20,000 weightings: what is settled vs. what is judgement



96.4%

of weightings put SVC-03 at the lowest priority, deprioritise with confidence.

65.1%

put SVC-06 top. Only SVC-04 or SVC-01 ever reach #1. A stable three-service cluster.

The boundary: both ends are act-on-able; ordering within the top cluster is a managerial judgement, not a settled result. Reproducible from seed 50 (≤ 0.011 on reseed).

FINDINGS

Answering the three research questions

RQ1

Yes — the framework itself.

Five operational indicators feed a TOPSIS model that orders services by debt priority with no recourse to source code. Its multi-criteria form is not decorative: it diverges from an incident-only view.

RQ2

Valid — to a bounded extent.

The composite behaves as an instrument should: differentiated profiles, no single indicator reproducing the order. Criterion validity is not claimed; the proxy link remains a proposition that survived every test this design allows.

RQ3

Stable at the extremes.

Two weightings agree in 2025 (τ = 0.867); across 20,000 draws SVC-03 is last in 96.4% and SVC-06 first in 65.1%. The top is a stable cluster; its internal order is a managerial call.

What it adds — and where it stops

Contributions

Empirical: to current knowledge, the first TOPSIS on live ITSM/CMDB records for TD prioritisation in financial trading IT.

Conceptual: an operationally-defined indicator set with an explicit literature-to-dimension mapping.

Methodological: the methods diverge on whether a single-dimension extreme should top the list. A compensatory choice; absolute-mode also adds rank-stability.

Practical: turns routine data into an auditable priority ordering; portfolio oversight; FINMA 2023/1 relevance.

Limitations — held in view

- Six-service demonstration set limits the power of the Kendall's τ tests.
- The same dataset builds and evaluates the framework → robustness & method-independence, not criterion validity.
- Cross-sectional snapshot; a service mid-remediation can look worse than it is.
- CFR shares its incident numerator with incident frequency; recording practices vary across teams.
- The single top rank stays weight-dependent within the SVC-06 / SVC-04 / SVC-01 cluster, a compensability judgement.

REFLECTION

A personal reflection

The hard part was definitional

I expected the difficulty to be computational. It was deciding what an incident count or a derived change failure rate can legitimately mean.

Disagreement taught the most

I expected the other methods to just confirm the order. Their split over SVC-06 forced an explicit stance: are debt dimensions compensatory?

Design Science set the standard of proof

It held the line between what the artefact demonstrates and what it merely suggests. Operational data can prioritise debt where source code cannot reach.

Where it goes next

1

External validation

Replicate at two or three further regulated institutions; compare weights and dimension mappings to test the proxy proposition beyond one case.

2

Longitudinal evaluation

A lagged design tracking whether high-priority services later generate predicted consequences, the route to criterion validity.

3

Portfolio-wide deployment

Replace the demonstration set with a live ranking; larger n also strengthens the stability tests.

4

Externally-anchored bounds

Derive absolute-mode bounds from regulatory / cross-industry end-of-support benchmarks, comparable across organisations.

The framework does not measure technical debt. *It orders services by the operational shadow debt casts: reproducibly, defensibly, from data the organisation already holds.*

Thank you, I welcome your questions. Tobias Zeier · Enterprise IT Management · University of Essex

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